DEMO PROJECT · AI AUTOMATION

An AI customer team that checks its own answers before customers see them

Supervisor + Sales, Support and Billing agents
built with n8n, Supabase, Stripe and LLMs

THE PROBLEM

Chatbots sound sure, even when they are wrong

Repetitive questions

Prices, order status, refunds and billing questions fill the day of a small team.

Invented answers

A basic chatbot can make up prices, discounts or refund promises.

No record

Nobody can see why the bot said what it said.

THE SOLUTION

A team of specialists, reviewed by a second AI

Sales agent

Answers from the real product catalogue. Quote totals are calculated by code, never by AI.

Support agent

Reads the customer’s own orders. Refunds and complaints become tickets for your team.

Billing agent

Reads usage, invoices and subscription from Stripe. Changes become tickets.

A Supervisor routes each message, then a second AI reviews the answer: approve, retry once, or send to a human.

SYSTEM ARCHITECTURE

From message to answer, with checks at every step

Customer message → Validate + identify → Classify intent → Recall memory

ROUTE TO A SPECIALIST (low confidence or human request goes straight to the human queue)

Sales agent
Product search (RAG), draft quotes. Reads products, writes quotes

Support agent
Own orders only. Complaints and refund requests become tickets

Billing agent
Stripe usage, invoices, subscription (read-only)

AI reviews AI (2nd LLM call) → Approved: reply + usage meter | Weak: retry once | Doubtful: human queue

supervisor_audit_log: every decision | customer_memory: per customer | Stripe meter: approved answers

TECHNOLOGY

Built from proven, replaceable parts

Orchestration

n8n workflows: one Supervisor and three separate specialist sub-workflows

Data and memory

Supabase Postgres with pgvector for product search and per-customer memory

AI

Groq-hosted LLM for classification, answers and review. Mistral embeddings.

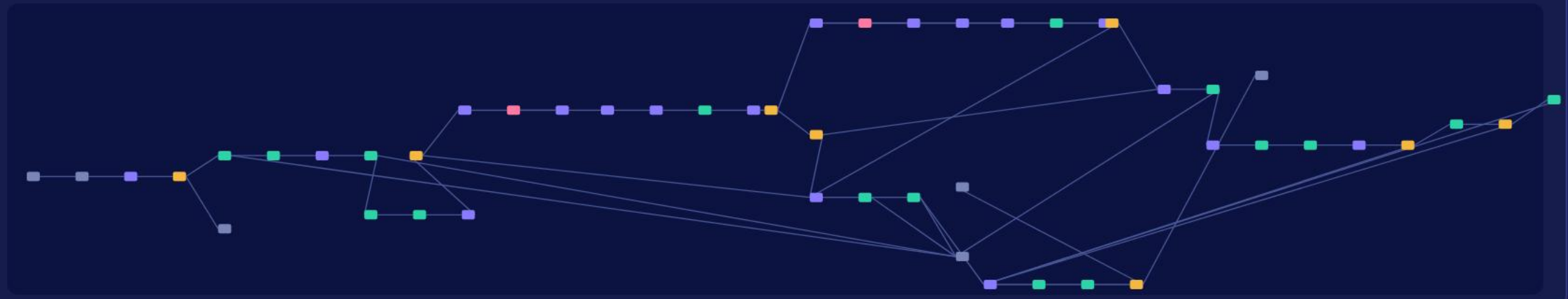
Billing

Stripe Billing Meters: usage events, subscription and invoices

REAL IMPLEMENTATION

Workflow maps drawn from the actual exports

Supervisor · 50 nodes



Sales agent · 17 nodes



Support agent · 16 nodes



Billing agent · 25 nodes



code · HTTP / database / API · decision · sub-workflow call · trigger / other

Generated from the exported workflow JSON (node positions and connections). A visualisation, not screenshots of n8n.

BUSINESS VALUE

What this approach gives a business

- **Answers come from your data**, or the system says it does not know
- **People handle only exceptions**: refunds, cancellations and doubtful cases go to a human
- **Every decision is logged** with intent, route, review result and outcome
- **Usage-based billing ready**: each approved answer is metered to Stripe
- **Modular**: each specialist is its own workflow, so one failing does not stop the others

Demo project built for portfolio purposes. Not a client deployment; no performance results are claimed.